



Implementation Infrastructure (II)

II Overview

Required PA Information

Intent

Ensures that the processes and assets important to an organization's performance are habitually and persistently followed, used, and improved.

Value

Sustains the ability to consistently achieve goals and objectives efficiently and effectively.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

II 1.1 Perform processes that address the intent of the Level 1 practices.

Level 2

II 2.1 Provide sufficient resources, funding, and training for developing and performing processes.

II 2.2 Develop and keep processes updated, and verify they are being followed.

Level 3

II 3.1 Use organizational processes and process assets to plan, manage, and perform the work.

II 3.2 Evaluate the adherence to and effectiveness of the organizational processes.

II 3.3 Contribute process-related information or process assets to the organization.

Level 4

II 4.1 Develop the organizational capability to understand and apply statistical and other quantitative techniques to accomplish the work.

Additional PA Explanatory Information

Establish the necessary infrastructure to ensure that processes are built, followed, sustained, and improved over time. The term “infrastructure” refers to everything needed to implement, perform, and sustain the organization’s set of processes. The infrastructure includes:

- Recorded processes to reflect how work is done
- Resources, e.g., people, tools, consumables, facilities
- Funding to perform the processes
- Training to perform the processes
- Objective evaluations to ensure that work is performed as intended

Implementation Infrastructure addresses an organization’s or project’s set of processes, not the model Practice Areas or practices. This verifies that processes are habitual and persistent, even during times of stress or change. Implement processes to meet the intent of these practices to improve process sustainment and integration throughout the organization.

Related Practice Areas

Governance (GOV)

Context Specific

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

The approach for managing data is supported by processes that define standards and guidelines for conducting activities necessary to achieve the relevant goals. Resources for managing data must be provided, and include personnel with defined roles and responsibilities, tools, and repositories for data and metadata. Processes and policies should cover the data lifecycle activities including data profiling, data minimization, data cleansing, data quality assessment, data retirement, and monitoring activities. These activities can be applied to areas such as data store consolidations, data warehousing, source to target transformation, data conversion, etc.

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

Robust infrastructure and processes are foundational to any DevSecOps implementation and necessary to enable continuous deployment. A properly configured infrastructure provides DevSecOps teams with speed, automation, efficiency, quality, and enhanced cybersecurity. Invest early in a properly designed DevSecOps infrastructure, and periodically evaluate the effectiveness and efficiency of the infrastructure and processes to make implementation more successful and sustainable. Infrastructure as Code (IaC) is an approach used to manage infrastructure and shift security and test left on the timeline.

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Consider and implement safety requirements within the infrastructure.

Examples of items to consider for processes, process assets, training, protocols, and resources:

- Eye wash stations
- First aid kits
- Physically disabled accessibility
- Staircase safety parameters
- Hazmat containers
- Defibrillators
- Training for safety equipment
- Evacuation and fire drills
- Active shooter drills
- Safety reports
- Hazard logs

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Develop a security strategy and related objectives, consistent with security policies, security guidelines, procedures, technical security measures, and tools; to protect the infrastructure where the work products are developed, stored, or delivered. Examples of work products that require protection include plans, source code, design documents, and solutions. When defining security objectives, consider confidentiality, integrity, and availability of the work products.

Areas to define include:

- Technical security measures like encryption or network protection
- Physical access control, e.g., locked rooms and physical badge access

- Procedures for account management, password handling, access control, handling of backups, etc.
- Policy for handling proprietary or confidential data

Ensure proper review and audit mechanisms are included within the security procedures to confirm the security controls are being appropriately and consistently applied.

Level 1

II 1.1

Required Practice Information

Practice Statement

Perform processes that address the intent of the Level 1 practices.

Value

Improves the likelihood that solutions are complete, correct, and timely.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Establish the fundamentals of process discipline needed to develop and deliver solutions efficiently and effectively to the customer. Solutions can be developed and provided without following a recorded process or plan. The discipline with which these practices are performed depends on the individuals managing and performing the work and can vary considerably.

An organization can successfully develop, deliver, and acquire solutions, even if:

- Processes are unrecorded, ad hoc, or chaotic
- Organizational infrastructure to support processes does not exist
- Success is based solely on competence and heroic efforts of people

However:

- Solutions may frequently exceed resource and schedule constraints
- Solutions do not consistently meet the customer's requirements
- Success may not be repeatable

Example Activities

Example Activities	Further Explanation
Perform processes.	

Example Work Products

Example Work Products	Further Explanation
Outputs of processes	Refer to example work products in the Level 1 practices.

Level 2

II 2.1

Required Practice Information

Practice Statement

Provide sufficient resources, funding, and training for developing and performing processes.

Value

Increases the likelihood of successful process improvement efforts by having sufficient resources.

Additional Required Information

Identify and provide adequate resources, funding, and training to individuals who own processes and to those who are identified to perform a process role in recorded project or organizational processes. Consider resource capacity and availability when allocating resources.

Explanatory Practice Information

Additional Explanatory Information

Sufficient resources for developing processes should consider:

- Leveraging provided funding
- Skilled people trained or experienced in:
 - Process development
 - Domain knowledge
 - Quality assurance
- Appropriate tools
- Training materials
- Time to perform the work

Develop a budget to support process activities in addition to work activities. Allocate funding for developing the processes, which may include:

- Recording and updating the process
- Acquiring or building tools
- Developing training materials
- Providing training
- Providing post-deployment support to process users
- Evaluating the processes

Train the people who are responsible for developing processes and performing quality assurance activities. Training supports successful process deployment by establishing a common understanding and providing skills and knowledge needed to perform the process.

Example Activities

Example Activities	Further Explanation
Identify and provide needed resources, as per capacity and availability needs.	Assign people with the needed skills and experience.
Determine budget.	
Develop or buy tools.	
Create or acquire training materials.	
Provide training.	

Example Work Products

Example Work Products	Further Explanation
Budget for resources	Verify the budget supports resource capacity and availability.
Training materials	
List of needed people, roles, and skills	
Tools	
Training records	

Related Practice Areas

Governance (GOV)

Organizational Training (OT)

Planning (PLAN)

Context Specific

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

DevSecOps teams require funding for tools and training to enable process automation from code commit through deployment. These tools enable the team to perform automated code builds; unit integration, security, and system testing; and establish a continuous integration and deployment environment. This environment is foundational to enabling the team to perform Continuous Delivery (CD) or Continuous Deployment. Continuous Development is often referenced together with Continuous Integration / Continuous Delivery (CI/CD). CI is the unification of individual components which occurs on a regular basis. Another basic infrastructure is Continuous Deployment, the automated transfer of software directly into a production environment. CD relies on rigorous static testing of source code and dynamic testing of deployable artifacts. DevSecOps teams receive training to gain an understanding of basic security practices, automated tools, continuous integration, environment configuration, and operations; and adopt and integrate these practices into their build and deployment processes.

Infrastructure as Code (IaC) is growing in application where code is written to build infrastructure such as networks and virtual machines. IaC is the managing and provisioning of infrastructure through code instead of through manual processes. Configuration files are developed that contain machine readable infrastructure specifications, making IaC the process of managing and provisioning computer data centers through machine-readable definition files, rather than physical hardware configuration or interactive configuration tools. It is easier to edit and distribute IaC configurations rather than physical hardware configuration or manually using configuration tools.

II 2.2

Required Practice Information

Practice Statement

Develop and keep processes updated, and verify they are being followed.

Value

Minimizes waste by ensuring affected stakeholders focus on the most valuable activities that are recorded in processes.

Additional Required Information

Include clear identification of process ownership and process roles within recorded processes. Monitor and verify how well processes are working and address any shortcomings or improvements.

Explanatory Practice Information

Additional Explanatory Information

Understanding and recording a process is the first step in specifying the way work is to be performed. A process can be recorded in many ways, but at a minimum includes a purpose, inputs, a sequence of steps or activities, outputs, roles, and responsibilities. This helps ensure repeatability and achievement of business and performance objectives. Verify that processes are being followed and kept up to date.

A common example of a process description is EITVOX, which includes:

- **E**ntry criteria
- **I**nputs
- **T**ask or activity descriptions
- **V**erification and validation
- **O**utputs
- **E**Xit criteria

Recorded processes provide support for consistent execution, repeatability of past successes, and an approach for learning and improving. A recorded process reduces risks that can impact quality, time-to-market, and customer satisfaction, including:

- Misunderstandings
- Unknown status
- Inconsistent performance
- Unavailable resources
- Unnecessary activities
- Missed activities
- Uncoordinated efforts
- Unclear roles and responsibilities
- Unachieved objectives

Example Activities

Example Activities	Further Explanation
Identify process purpose.	
Determine process description format.	
Describe and record processes.	
Perform processes.	
Verify that processes are being followed.	Multiple techniques can be used, such as: • Objective evaluations • Audits • Retrospectives • Process reviews • Peer reviews
Review and update recorded processes with affected stakeholders.	
Communicate and make recorded processes available.	This includes changes to existing processes as well as new processes.

Example Work Products

Example Work Products	Further Explanation
Recorded processes	
Process verification results	

Related Practice Areas

Governance (GOV)

Peer Reviews (PR)

Process Asset Development (PAD)

Process Quality Assurance (PQA)

Context Specific

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

DevSecOps teams make extensive use of measures and use tools to develop dashboards that enable the teams to monitor their performance, verify processes are being followed, and to evaluate and improve the overall health of their environments and pipelines. Review impacted processes when new tools are adopted and periodically for consistency. Analyzing verification results, process adherence, and corresponding measurements enable the organization to target what process improvements should be made. Retrospectives are an example of where teams review their processes for their effectiveness.

Level 3

II 3.1

Required Practice Information

Practice Statement

Use organizational processes and process assets to plan, manage, and perform the work.

Value

Leverages organizational learning and use of best practices, leading to reductions in rework and cost, and increases in efficiency and effectiveness.

Additional Required Information

Organizational processes and process assets must include clear ownership and identification of processes and process roles.

Explanatory Practice Information

Additional Explanatory Information

Organizational process assets enable consistent process execution across the organization.

The availability and appropriate use of consistent organizational process assets help to:

- Perform planning and work activities based on proven practices
- Facilitate the transfer of personnel to where needs are most critical
- Reduce the likelihood of repeating issues and mistakes
- Apply process assets to provide the most benefit to projects and performance

Organizations typically keep updated a set of standard process assets to be used and tailored depending on the type of work performed. For example, an organization may have different planning processes for various types of work.

Example Activities

Example Activities	Further Explanation
Plan work using organizational process assets.	Tailoring of assets typically occurs during planning.
Manage work using organizational process assets.	
Perform work following organizational process assets.	

Example Work Products

Example Work Products	Further Explanation
Tailored process assets	May include: • Templates specific for projects • Processes and procedures specific for the project • Lifecycle models specific for the project • Operational definitions of measures specific for the project • Checklists specific to a type of work
Work products resulting from using the process assets	

Context Specific

People

Context Tag:	CMMI-PPL
Context:	Use processes to align workforce performance and business objectives.

Develop processes consistent with workforce competency descriptions. Workforce competencies are strong influencers on process performance and should be considered for the most critical processes.

II 3.2

Required Practice Information

Practice Statement

Evaluate the adherence to and effectiveness of the organizational processes.

Value

Provides insight on potential cost-effective improvements to organizational processes and how they are used.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Evaluating adherence helps ensure that processes and process assets are understood, relevant, effective, and used as intended.

Assessing effectiveness of processes and process assets helps keep them relevant to the business needs and strategy. Analyze processes and process assets periodically. This analysis helps to understand their strengths and weaknesses and to continually improve them to provide value to the organization.

Methods for evaluating adherence and effectiveness include:

- Observation
- Evaluations, assessments, or audits
- Interviews
- Analysis of the use of work products and results

Effectiveness includes:

- Ease of use
- Fewer mistakes
- Optimal use of resources
- Timely delivery
- Improved performance
- Increased customer satisfaction
- Capability that meets needs

When assessing effectiveness, some questions that may help are:

- Why are we doing this?
- Who is the target audience?
- Can this be done in a simpler way?
- What is working?
- What is not working?

Process performance measurements can also be used to analyze the effectiveness of a process.

The benefits of a process can be demonstrated by performance improvements such as:

- Meeting objectives
- Reducing costs
- Reducing defects
- Increasing productivity
- Reducing cycle time
- Increasing customer satisfaction
- Increasing market share

Example Activities

Example Activities	Further Explanation
Evaluate processes for effectiveness and usefulness.	
Analyze process performance measurement results.	
Examine results of evaluations, appraisals, or audits.	

Example Activities	Further Explanation
Record and communicate results to affected stakeholders.	
Submit improvement proposals.	

Example Work Products

Example Work Products	Further Explanation
Evaluation results	Include: • Organizational best practices • Reusable process assets • Issues, including: ○ Non-compliance ○ Effectiveness ○ Behavioral • Recommendations and opportunities for improvement • List of major risks associated with process implementation
Analysis results	Include: • Process performance • Trends in customer responses ○ Satisfaction ○ Complaints • Solution reliability • Defect insertion rate • Cycle time • Solution quality • Root causes
Improvement proposals	

Related Practice Areas

Causal Analysis and Resolution (CAR)

Managing Performance and Measurement (MPM)

Process Quality Assurance (PQA)

II 3.3

Required Practice Information

Practice Statement

Contribute process-related information or process assets to the organization.

Value

Increases return on investment by improving the organizational processes and process assets.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Contributing and using process-related information helps keep processes and process assets relevant, current, and aligned with organizational needs. Users and implementers of processes should contribute process use and improvement information to the organization and be involved in determining what information will be kept and used.

Ensure:

- Projects systematically contribute to organizational learning
- Key intellectual capital does not leave the organization when people leave
- Future projects can draw from existing process assets to become productive quickly

Collect process-related experiences such as:

- Best practices
- Work products
- Measures
- Lessons learned
- Process improvement proposals

Store measurement data in the organization's measurement repository and other information in the organization's Process Asset Library (PAL). Make these available to those planning and performing similar processes.

Example Activities

Example Activities	Further Explanation
Collect and record best practices, lessons learned, and information from tailoring the processes.	
Submit assets for potential inclusion in the organization's process asset library.	Include the context along with assets to make them more useful.

Example Activities	Further Explanation
Propose improvements to the organizational process assets.	

Example Work Products

Example Work Products	Further Explanation
Examples, best practices, and lessons learned	Include assets that: • Worked well • Are ineffective • Could be improved
Tailoring records, rationale, worksheets, and other related work products associated with tailoring and implementing the organization's set of standard processes for the work	If multiple projects tailor the same aspects of the process, this is an indication that the process may need to be updated.

Related Practice Areas

Managing Performance and Measurement (MPM)

Process Asset Development (PAD)

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Consider the following sources of information for contributing safety process-related information or process assets to the organization:

- Safety logs
- Safety compliance reports
- Results of applying new safety methods, tools, and controls
- Stakeholder requests and information
- Experiences from performing safety training
- Test results from safety drills

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Consider the following sources of information for contributing security process-related information or process assets to the organization:

- Discovered threats
- Reusable security solution risk mitigations
- Security project data, e.g., security lessons learned, security-related effort and cost, and measurements
- Results of applying new security methods, tools, and controls
- Security architecture and design standards
- Security technology evaluation criteria
- Security design analysis results
- Security configuration standards
- Security solution verification and validation results and reports
- Experiences from performing security training
- Identified vulnerabilities
- Incident resolution approaches
- Results from threat intelligence analysis
- Stakeholder requests and information

Level 4

II 4.1

Required Practice Information

Practice Statement

Develop the organizational capability to understand and apply statistical and other quantitative techniques to accomplish the work.

Value

Empowers the workforce to use information to make effective changes and meet or exceed business objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

As business objectives and Quality and Process Performance Objectives (QPPOs) evolve, the organizational capability to process and use information must keep pace to ensure effective support from the workforce. This in turn requires deliberate strategic updates to the training, skills and ability needs of the organization in the use of statistical and other quantitative techniques.

The development of the organizational capability to understand statistical and other quantitative techniques incorporates the following aspects:

- Training individuals to obtain relevant knowledge and skills for statistical and other quantitative techniques
- Providing the trained individuals with the opportunity to develop their skills by applying the techniques
- Ensuring the trained and skilled individuals have the infrastructure to perform the techniques according to defined processes

Example Activities

Example Activities	Further Explanation
Define infrastructure required to support the use of statistical and quantitative techniques.	Typically include: • Trained resources and funding for establishing an organizational function for conducting statistical and quantitative management activities • Statistical tools, e.g., Minitab, QI Macros

Example Activities	Further Explanation
Develop training materials and job aids for using statistical and quantitative techniques.	
Monitor effectiveness of the organizational capability.	
Make improvements to the organizational capability.	Review and consider improvements to ensure that updates to statistical and other quantitative techniques reflect: • Changes in business objectives • Changes in QPPOs • Skill maturity of the workforce • Latest industry advances and trends

Example Work Products

Example Work Products	Further Explanation
Requirements of organizational capability for using statistical and quantitative techniques	
Statistical and quantitative training materials	
List of improvements	Prioritize improvements based on business objectives, considerations for competitive advantage, and workforce skills and knowledge.

Related Practice Area

Managing Performance and Measurement (MPM)

Organizational Training (OT)

